

# Precision, Accuracy, and Geographic Irreplaceability of Volunteer Chloride Monitoring: Apparent Observer Bias in Oklahoma's Blue Thumb Data Is a Geographic Artifact

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## ABSTRACT

Citizen science programs collect environmental data at scales impossible for professional agencies alone, yet a persistent trust gap limits their use, often because “volunteer” is treated as a categorical proxy for lower reliability. The stakes of this distrust are concrete: 93% of the 327 Blue Thumb chloride monitoring sites in Oklahoma have no professional monitor within 1 km, so discarding volunteer data on principle leaves most of the monitored landscape unvalidated. We assess Blue Thumb volunteer chloride data, which the Oklahoma Conservation Commission maintains separately from its professional records, enabling programmatic separation without new field sampling. Variance decomposition of 2,566 volunteer titration replicate pairs shows that volunteers are highly repeatable within the method's 5 mg/L resolution: 97.3% of replicate pairs agree within one drop. The apparent advantage in volunteer repeatability (mean  $|\text{diff}| = 1.53$  vs. 2.51 mg/L for professionals) reflects the quantization floor of the drop-count titration, not greater observer consistency. Eighteen years of indoor quality assurance (867 tests against known standard solutions) show volunteers read slightly high (+4 to +5 mg/L at concentrations at or below 100 mg/L), ruling out systematic under-counting. A regional mixed-effects model ( $N = 895$  observations, 92 sites) indicated an apparent systematic low bias ( $\beta_{\text{IsVolunteer}} = -0.433$ ,  $p = 0.047$ ), but volunteer and professional sites in that model have zero geographic overlap. Stratified random subsampling to balance the documented West–East salinity gradient (Dyer et al., 2025) eliminated the effect entirely ( $p = 0.54$ ), establishing the apparent bias as a geographic artifact rather than observer error. We conclude that Blue Thumb volunteer chloride data is precise within its resolution limit, accurate against known standards, and geographically irreplaceable, and that Oklahoma's separable data infrastructure offers a replicable model for any program that maintains volunteer and professional records as distinct, queryable datasets.

Keywords: citizen science, water quality, chloride, validation, volunteer monitoring, mixed-effects model, EPA Water Quality Portal, Blue Thumb

## 1 INTRODUCTION

### 1.1 The citizen science data quality problem

Most of the monitored stream landscape depends on volunteers. In Oklahoma, 93% of the 327 Blue Thumb chloride monitoring sites have no professional monitor within 1 km, so a decision to discard volunteer data on principle is a decision to leave most of the monitored landscape unvalidated. This is the stake that motivates the present study. Yet a persistent “trust gap” separates volunteer-collected data from formal scientific and regulatory use. The core objection is not that volunteers are careless, but that their instruments are less precise, their training less standardized, and their measurements therefore less comparable to professional reference data. This skepticism is often applied categorically: the word “volunteer” functions as a proxy for lower reliability regardless of the specific program's protocols, quality assurance practices, or track record (Metcalf et al., 2022).

A recent review of 72 citizen science water quality studies found that while volunteer-collected data

can align with professional measurements, concerns about precision, training variability, and method comparability persist as barriers to regulatory acceptance (Blanco Ramirez et al., 2023). These concerns are well-documented across multiple dimensions: data quality assurance requirements, analytical method standardization, and institutional trust (San Llorente Capdevila et al., 2020). A systematic review of existing volunteer water monitoring datasets concluded that the majority of comparison studies report volunteer data of comparable quality to professional data, yet many existing datasets remain underutilized (Albus et al., 2019). Participatory science involvement in water quality governance remains limited by gaps between data collection capacity and regulatory application (O’Ryan et al., 2025).

Most validation studies attempting to address this gap rely on resource-intensive, co-located side-by-side sampling: a professional technician and a volunteer measure the same water body simultaneously, and the paired results are compared. This prospective design, while rigorous, is expensive, spatially limited to the sites where both parties are present, and temporally constrained to the dates of the coordinated sampling events. It cannot capture the long-term performance characteristics of a monitoring program across its full geographic footprint. A retrospective approach, comparing years of archival volunteer data against years of archival professional data from the same streams, would be far more powerful, but is typically impossible because volunteer and professional records are not programmatically distinguishable in national data repositories.

Published examples of this approach include co-located field kit versus laboratory comparisons for multiple water quality parameters (Quinlivan et al., 2020), parallel citizen science and government monitoring programs evaluated over multi-year periods (Dickson et al., 2024), and long-term comparisons of citizen science water quality indices against state agency records (de Camargo Reis et al., 2026). While these studies demonstrate that co-located validation is feasible and generally supports volunteer data quality, they share a common constraint: the validation is limited to the sites and dates where coordinated sampling occurred.

## 1.2 Why Oklahoma is uniquely positioned for retrospective validation

The EPA Water Quality Portal (WQP) aggregates water quality monitoring data from state, federal, tribal, and local agencies across the United States. Each contributing organization submits data under a unique OrganizationIdentifier, which is preserved in the WQP record and queryable through the public API. In principle, this metadata field allows volunteer and professional records to be separated by organization code, but only if the contributing organization assigns a distinct code for each monitoring program.

In practice, most major volunteer programs do not submit their data to the WQP at all. A query of the WQP organization registry (1,265 U.S. contributing organizations, retrieved June 2026) returns no organization identifier for two of the most nationally recognized volunteer water quality programs, Missouri Stream Team and Georgia Adopt-A-Stream; neither submits its volunteer monitoring data to the WQP. Both instead maintain volunteer records in separate, program-specific databases (the Missouri Department of Conservation Stream Team Portal and the Georgia Adopt-A-Stream database, respectively). This is not a limitation of the WQP, which does support distinct volunteer organization codes: 141 U.S. organizations are flagged as volunteer programs in the registry. These two programs are simply not among them, so the retrospective comparison approach used here cannot be applied to them.

Oklahoma’s Blue Thumb program is the enabling exception. Like the programs above, the Oklahoma Conservation Commission (OCC) does not place volunteer data in the WQP; the professional Rotating Basin data is submitted under the organization identifier `OKCONCOM.WQX`, while the volunteer data is maintained in a separate but independently and programmatically accessible OCC system: an R-Shiny portal and a public ArcGIS REST API. This infrastructure enables, for the first time, a large-scale retrospective validation of volunteer water quality measurements against professional reference data without requiring new field sampling, co-located visits, or any cost beyond computational resources.

**A critical note on data identity:** Early analysis in this study incorrectly treated `OKCONCOM.WQX` data in the EPA WQP as Blue Thumb volunteer data. This organization code in fact represents the OCC Rotating Basin professional monitoring program (Method 9056). The error was identified by OCC Quality Assurance Officer Kim Shaw in January 2026. All results presented in this paper use the corrected data source: a separate OCC R-Shiny export of verified Blue Thumb volunteer records, SHA-256 hash-verified at runtime, and assigned the internal identifier `BLUETHUMB_VOL` in the analysis pipeline (the runtime enforcement of this distinction is described in the Reproducibility subsection). The corrected volunteer dataset was independently cross-validated against OCC’s live ArcGIS production database (2,026 of 2,027

98 overlapping records match exactly at the WBID+date level).

### 99 **1.3 Chloride as a validation target**

100 Chloride is a conservative ion: it does not precipitate, adsorb to sediments, volatilize, or participate in  
101 biotic uptake under typical freshwater conditions. This chemical stability makes chloride one of the  
102 most reliable tracers of anthropogenic water quality impacts and, critically for this study, one of the best  
103 analytes for method comparison work. A concentration difference between volunteer and professional  
104 measurements of chloride reflects the measurement methodology, not environmental transformation of  
105 the analyte between sampling events.

106 Chloride has been established as a primary indicator of freshwater salinization syndrome, a coupled  
107 process of rising dissolved salts, alkalization, and contaminant mobilization that has affected 37% of the  
108 drainage area of the contiguous United States over the past century (Kaushal et al., 2018). Its chemical  
109 stability in freshwater systems makes it one of the most reliable tracers for distinguishing dilution and  
110 evaporation from chemical transformation (Hem, 1985).

111 Elevated chloride in Oklahoma streams is associated with oil and gas produced water, road salt  
112 application, agricultural irrigation return flow, and wastewater effluent. Background concentrations  
113 vary substantially across the state, with a documented West–East gradient driven by geology: western  
114 Oklahoma’s Permian Basin evaporites produce naturally saline groundwater that feeds surface streams,  
115 while eastern Oklahoma streams are naturally fresher. This gradient has been characterized by Dyer  
116 et al. (2025), who used natural landscape and instream habitat features to classify Oklahoma streams into  
117 distinct geochemical groups. The West–East salinity gradient is directly relevant to this study’s geographic  
118 confound analysis and is discussed in detail in Section 3.4.

119 The Silver Nitrate titration method used by Blue Thumb volunteers measures chloride through  
120 a precipitation reaction (the argentometric method, Standard Methods 4500-Cl B) (American Public  
121 Health Association et al., 2017): chloride ions in the water sample react with silver nitrate solution,  
122 and the endpoint is detected visually by a color change from yellow to brick-red. Each drop of titrant  
123 corresponds to 5 mg/L chloride at the standard dilution factor, imposing a quantization floor on all  
124 volunteer measurements. Volunteer chloride readings are exact multiples of 5 mg/L by construction. This  
125 quantization is not a flaw in volunteer execution; it is an inherent property of the analytical method and is  
126 analyzed explicitly in Section 3.1.

### 127 **1.4 The Blue Thumb program**

128 Blue Thumb is an Oklahoma Conservation Commission citizen science program that has trained and  
129 deployed volunteers to monitor Oklahoma stream reaches since 1992. The program was designed with  
130 protocols that deliberately mirror the OCC’s professional water quality sampling standards, as documented  
131 in the OCC Quality Assurance Project Plan (QAPP; Bond, 2026). The program operates under a Quality  
132 Management Plan and has been funded through the EPA’s Section 319 Program targeting nonpoint source  
133 pollution. Key program features relevant to this study include:

- 134 • Volunteer training includes both classroom and field components before independent monitoring  
135 begins.
- 136 • OCC water quality specialists conduct field monitoring alongside volunteers on a rotating basis,  
137 providing direct quality oversight.
- 138 • A formal indoor quality assurance program has been in continuous operation since at least 2007, in  
139 which volunteers test their chloride kits against solutions of known concentration under controlled  
140 laboratory conditions.
- 141 • Field QA sessions assess additional parameters (temperature, Secchi depth, sample collection  
142 procedure) on a rotating basis.
- 143 • Data are submitted to OCC’s ArcGIS database (via Survey123) and cross-referenced against a  
144 separate R-Shiny export portal.
- 145 • Volunteer chemical data is submitted for upload into the EPA’s STORET database, making it  
146 available to outside researchers and agencies.

- 147 • Volunteer monitoring and education hours are valued at over \$30.00 per hour and contribute to  
148 matching federal funds with in-kind services.

149 The program has collected data from over 300 streams across Oklahoma, with over 100 sites currently  
150 active. Critically, the program monitors reaches that professional agencies do not: 93% of Blue Thumb's  
151 327 chloride monitoring sites are more than 1 km from the nearest professional monitoring station in  
152 the EPA WQP. This geographic complementarity is central to the program's value proposition and is  
153 examined quantitatively in Section 3.5.

## 154 1.5 Gap in the literature

155 Despite more than 20 years of monitoring records and a documented QA framework that mirrors profes-  
156 sional standards, Blue Thumb chloride data has not been subjected to published mathematical validation  
157 against professional agency measurements. More broadly, no published study has exploited the EPA  
158 WQP's OrganizationIdentifier infrastructure as the basis for large-scale retrospective citizen science  
159 validation. This represents both a gap in the literature and a missed opportunity: if the retrospective  
160 approach is valid for Oklahoma's Blue Thumb program, it could be applied to any volunteer monitoring  
161 program that submits data to the WQP with a distinct organization code, providing a zero-cost, scalable  
162 template for citizen science data validation nationally.

163 The Water Quality Portal, developed by the EPA, USGS, and National Water Quality Monitoring  
164 Council, is the largest standardized water quality dataset available, with over 290 million records from  
165 more than 2.7 million sites (Read et al., 2017). Despite this infrastructure, a systematic search of the  
166 literature found no published study that has exploited the WQP's OrganizationIdentifier metadata to  
167 programmatically separate citizen science from professional records for retrospective method validation.  
168 Albus et al. (2019) identified only six studies that used archival datasets for volunteer-professional  
169 comparison, none of which employed systematic spatial-temporal matching criteria or leveraged the  
170 WQP's organizational metadata structure.

## 171 1.6 Study objectives

172 This study addresses three specific objectives:

- 173 1. **Precision:** Quantify volunteer within-test repeatability using titration replicate pairs, and contextu-  
174 alize this precision within the known 5 mg/L quantization floor of the Silver Nitrate method.
- 175 2. **Accuracy:** Assess volunteer absolute accuracy using 18 years of historical indoor QA records in  
176 which volunteer kits were tested against solutions of known chloride concentration.
- 177 3. **Confound identification:** Identify and account for the geographic confound that could artifactually  
178 inflate or deflate apparent volunteer bias in a regional comparison of volunteer and professional  
179 measurements.

# 180 2 METHODS

## 181 2.1 Data sources

182 **Volunteer data provenance.** The R-Shiny export was provided by OCC and SHA-256 hash-verified  
183 before each pipeline execution (hash: 56d21f...b206cd). The export was independently cross-  
184 validated against OCC's live ArcGIS FeatureServer database (bluethumb\_oct2020\_view): of 2,027  
185 records with overlapping WBID+date combinations, 2,026 match exactly (99.95%). The one discrepancy  
186 was a known data quality incident at Wolf Creek (2026-01-02) in which the Survey123 mobile application  
187 stored a zero rather than the correct titration value. The chloride field used is Chloride\_Low1\_Final,  
188 calculated as raw drop count multiplied by 5 mg/L with no blank subtraction.

189 **Professional data provenance.** Professional chloride measurements were downloaded from the EPA  
190 WQP using the public Results API, filtered to Oklahoma, CharacteristicName = Chloride, STORET and  
191 NWIS providers, dataProfile = resultPhysChem. Records with invalid coordinates, non-detect result flags,  
192 or negative concentration values were removed. Professional organizations included in the reference  
193 dataset are: OKWRB-STREAMS\_WQX (Oklahoma Water Resources Board), CNENVSER (Chickasaw  
194 Nation Environmental Services), O\_MTRIBE\_WQX (Otoe-Missouria Tribe), and additional tribal and state  
195 agencies (23 organizations total). The OCC Rotating Basin program (OKCONCOM\_WQX, Method 9056) is

**Table 1.** Data sources used in this study.

| Source                        | Description                                                                                                                        | Records                                     | Date range               |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------------------------|
| EPA Water Quality Portal      | Professional chloride measurements, Oklahoma streams (STORET + NWIS providers)                                                     | 18,299 records after cleaning               | 1993–2025                |
| OCC R-Shiny export            | Blue Thumb volunteer chloride measurements                                                                                         | 10,800 numeric values from 327 unique sites | 2005-01-04 to 2024-12-31 |
| OCC Rotating Basin QA dataset | Churn splitter duplicates ( $N = 113$ pairs), spatial replicates $\sim 100$ m apart ( $N = 114$ pairs), field blanks ( $N = 111$ ) | 338 total records                           | 2022-07-18 to 2024-05-21 |
| Blue Thumb indoor QA records  | Volunteers testing kits against known standard solutions                                                                           | 867 tests across 13 sessions                | 2007–2025                |

the source of the professional QA duplicates used in the precision analysis (Section 2.3) and contributes professional observations to the regional mixed-effects model.

**Rotating Basin QA dataset.** Provided by OCC statistician Joseph Dyer (via Karla Spinner, OCC) on February 20, 2026. Contains three record types distinguished by the Type field: Routine Sample and Field Duplicate (same grab sample analyzed separately, yielding within-test churn splitter duplicate pairs), Field Replicate (separate grab sample collected at the same site within the same day  $\sim 100$  m apart, yielding within-site spatial replicate pairs), and Field Blank (deionized water, yielding single contamination-check measurements). Samples that failed the Field Blank test were pre-excluded by OCC before delivery.

**Indoor QA dataset.** Provided by Kim Shaw (OCC) on March 2, 2026 (Shaw, 2026). Covers 14 quarterly sessions in which Blue Thumb volunteers tested chloride kits against solutions of known concentration under controlled indoor conditions. Spring 2013 was excluded from analysis because the standard solution was compromised during preparation (documented in Kim Shaw’s session notes). Thirteen sessions remain (2007–2025), yielding 867 individual tests.

## 2.2 Statistical analysis: regional model and geographic confound

**Regional Linear Mixed-Effects model.** The primary field model uses all volunteer and professional records within the QA dataset’s spatiotemporal window (2022–2024, north-central Oklahoma,  $N = 895$  observations, 92 sites). The temporal scope is restricted to the Rotating Basin QA dataset’s collection period (July 2022 to May 2024) so that the precision baselines established in Section 2.3 are contemporaneous with the field observations being modeled:

$$\log(\text{Chloride}) \sim \text{IsVolunteer} + \sin\left(\frac{2\pi \cdot \text{Julian}}{365}\right) + \cos\left(\frac{2\pi \cdot \text{Julian}}{365}\right) + \text{Longitude} + (1|\text{Site}) \quad (1)$$

This linear mixed-effects specification (Pinheiro and Bates, 2000; Zuur et al., 2009; Bolker et al., 2009) follows J. Dyer’s published framework for mixed-effects analysis of Oklahoma stream data (Dyer et al., 2025). REML estimation is used. The IsVolunteer term is the fixed treatment factor; seasonal harmonics (sine and cosine of Julian day) control for within-year chloride cycling, Longitude controls for the West–East salinity gradient, and a site-level random intercept absorbs persistent between-site differences. This model must be interpreted with caution: the 80 volunteer sites and 12 professional sites in the dataset have zero geographic overlap, so the IsVolunteer coefficient is inseparable from site-level differences and is not interpretable as a pure observer effect. This identifiability limitation is precisely what the stratified subsampling test below is designed to address.

**Geographic confound test: stratified random subsampling.** To test the hypothesis that the apparent volunteer low bias reflects the East–West geographic distribution of monitoring sites rather than observer error:

1. Compute the longitude distribution of all volunteer sites and all professional sites in the regional dataset.

- 229 2. Divide the volunteer longitude range into quartiles.
- 230 3. Randomly select 3 volunteer sites per quartile (12 total) to match the 12 professional sites in the  
231 dataset, balancing the geographic distribution.
- 232 4. Re-run the LME on the geographically balanced subsample using the same model specification.
- 233 5. Compare the IsVolunteer  $p$ -value to the full (unbalanced) model.

234 The subsampling was designed to test a specific, pre-specified hypothesis (geographic confound)  
235 rather than to optimize model fit. The hypothesis was proposed by J. Dyer prior to the subsampling  
236 analysis based on his published characterization of Oklahoma's West–East salinity gradient (Dyer et al.,  
237 2025).

### 238 2.3 Variance decomposition (precision analysis)

239 Using the Rotating Basin QA dataset and ArcGIS volunteer replicate data:

- 240 • **Volunteer within-test precision:** Absolute difference between paired titration replicates (test3 vs.  
241 test5 fields in the ArcGIS volunteer dataset).  $N = 2,566$  pairs. Drop-count differences converted to  
242 mg/L by multiplying by 5.
- 243 • **Professional within-test precision:** Absolute difference between churn splitter duplicate pairs  
244 from the same grab sample.  $N = 113$  pairs.
- 245 • **Professional within-site precision:** Absolute difference between spatial replicate pairs collected at  
246 the same site within the same day,  $\sim 100$  m apart.  $N = 114$  pairs.
- 247 • **Field blank contamination:** Single measurements of deionized water.  $N = 111$ . All values should  
248 fall at or below the detection limit.

249 The quantization effect is analyzed separately: the frequency distribution of volunteer within-test  
250 differences is expected to show a strong spike at zero (identical readings) due to the 5 mg/L resolution  
251 floor, with subsequent spikes at 5 mg/L intervals.

### 252 2.4 Indoor QA accuracy analysis

253 Using the 13-session historical indoor QA dataset ( $N = 867$  individual tests):

- 254 • Error computed as: (volunteer reading – known standard concentration) for each test.
- 255 • Results stratified by concentration range: low ( $\leq 50$  mg/L), mid (50 to 100 mg/L), high ( $>$   
256 100 mg/L).
- 257 • Mean error and direction of bias reported by concentration tier.
- 258 • Spring 2013 excluded due to compromised standard solution (K. Shaw, personal communication,  
259 March 2026).

260 The direction of the mean error is of particular interpretive importance: a systematic high bias rules  
261 out systematic under-counting as a kit accuracy failure mode, independent of any field comparison.

### 262 2.5 Geographic coverage analysis

263 For each of the 327 unique Blue Thumb chloride monitoring sites, the Haversine great-circle distance to  
264 the nearest professional monitoring station in the full EPA WQP Oklahoma dataset was computed. The  
265 fraction of volunteer sites beyond threshold distances (1 km, 5 km, 10 km, 25 km, 50 km) is reported.

## 2.6 Reproducibility

The analysis is implemented in Python 3.11 and is publicly available. The results reported here, the variance decomposition (Section 2.3), the indoor QA accuracy analysis, the regional mixed-effects model, and the stratified-subsampling confound test, are produced by the Phase 2 analysis scripts. Running the pipeline writes a `metadata.json` manifest (git commit hash and SHA-256 hashes of the inputs); the committed, continuous-integration-enforced reproducibility check is `verify.py`, which recomputes every headline number from the committed data (claim-by-claim mapping in `CLAIMS.md`). The volunteer data file is SHA-256 verified at runtime; the pipeline terminates with an explicit error if the file is absent or modified, with no silent fallback, preventing reuse of the incorrect WQP OKCONCOM.WQX data in place of the verified volunteer export.

## 3 RESULTS

Results are presented in the following order: precision and accuracy first (establishing what the data is capable of), then the regional mixed-effects model that surfaces an apparent observer bias together with the identifiability caution that the model cannot separate observer from site, then the stratified-subsampling test that resolves that confound as the primary result, and finally the geographic coverage that establishes why the volunteer data is irreplaceable.

### 3.1 Volunteer precision: variance decomposition

**Table 2.** Within-source variability across measurement types.

| Source                   | <i>N</i> | Mean  diff | 95% CI | Median  diff | Notes                                  |
|--------------------------|----------|------------|--------|--------------|----------------------------------------|
| Volunteer within-test    | 2,566    | 1.53 mg/L  | ±0.11  | 0.00 mg/L    | Titration replicates (test3 vs. test5) |
| Professional within-test | 113      | 2.51 mg/L  | ±0.72  | 1.00 mg/L    | Churn splitter duplicates              |
| Professional within-site | 114      | 3.58 mg/L  | ±1.10  | 1.00 mg/L    | Spatial replicates, ~100 m             |
| Field blanks             | 111      | 0.5 mg/L   | n/a    | n/a          | Deionized water; all below detection   |

The apparent superiority of volunteer within-test repeatability (mean |diff| = 1.53 mg/L vs. 2.51 mg/L for professionals) is a quantization artifact and must not be interpreted as greater observer precision. Because the Silver Nitrate titration resolves to 5 mg/L per drop, volunteers cannot detect concentration differences smaller than one drop. The distribution of within-test differences confirms this mechanism: 73.2% of volunteer replicate pairs are identical (zero difference), 24.1% differ by exactly 5 mg/L (one drop), and only 2.7% differ by 10 mg/L or more (two or more drops). In total, 97.3% of pairs agree within one drop. Professional instruments resolve to 0.1 mg/L and genuinely detect the sub-milligram variation that the volunteer method cannot resolve. The volunteer repeatability figure reflects the floor of what the drop-count method can distinguish, not observer consistency per se.

The field blanks ( $N = 111$  measurements of deionized water) returned 0.5 mg/L or lower in all cases, confirming that contamination from the sampling equipment is not a meaningful source of error (Figure 1A).

The quantization mechanism is visualized directly in Figure 2, which compares the distribution of chloride values measured by professional instruments (continuous, right-skewed) against volunteer measurements (discrete spikes at exact multiples of 5 mg/L). 100% of volunteer readings are quantized to 5 mg/L steps.

### 3.2 Volunteer accuracy against known standards

Thirteen indoor QA sessions spanning 2007 to 2025 (867 individual tests) provide the cleanest possible assessment of kit accuracy: no stream variability, no temporal matching uncertainty, and no geographic confounds.

Volunteers consistently read high across all concentration ranges. At normal operational concentrations ( $\leq 100$  mg/L, covering 603 of 867 tests and the vast majority of Blue Thumb monitoring sites), mean

**Table 3.** Volunteer accuracy against known standard solutions by concentration range.

| Concentration range   | <i>N</i> | Mean error  | SD         | Median error | Direction |
|-----------------------|----------|-------------|------------|--------------|-----------|
| Low ( $\leq 50$ mg/L) | 406      | +4.37 mg/L  | 4.67 mg/L  | +3.00 mg/L   | High      |
| Mid (50–100 mg/L)     | 197      | +5.32 mg/L  | 9.57 mg/L  | +5.00 mg/L   | High      |
| High ( $> 100$ mg/L)  | 264      | +32.20 mg/L | 26.30 mg/L | +40.00 mg/L  | High      |

error is within one drop (+4 to +5 mg/L), which falls within the method’s stated resolution. The standard deviation at low concentrations (4.67 mg/L) is comparable to a single drop, indicating tight clustering. At mid-range concentrations the SD increases to 9.57 mg/L (~two drops), and at high concentrations the SD reaches 26.30 mg/L, reflecting the compounding endpoint subjectivity described below. The increasing bias at elevated concentrations (mean +32 mg/L above 100 mg/L) is consistent with compounding endpoint subjectivity: each additional drop of titrant required for a high-concentration sample introduces another opportunity for endpoint misidentification.

The direction of the bias carries critical interpretive weight: volunteers read high, not low. Eighteen years of QA against known standards therefore rule out systematic under-counting as a kit accuracy problem; the kit, if anything, errs toward over-reporting. Figure 3 visualizes the error distribution and concentration-dependent scaling.

### 3.3 Regional mixed-effects model: an apparent bias that cannot identify its source

The regional mixed-effects model uses all volunteer and professional records within the QA dataset’s spatiotemporal window (2022–2024, north-central Oklahoma) to ask whether volunteer and professional readings differ at landscape scale across all 92 sites. It produces the following fixed effects:

**Table 4.** Regional GLMM fixed effects (REML estimation,  $N = 895$  observations, 92 sites).

| Fixed Effect   | Estimate ( $\beta$ ) | SE    | <i>p</i> -value | 95% CI           |
|----------------|----------------------|-------|-----------------|------------------|
| Intercept      | −37.663              | 7.640 | $< 0.001$       | [−52.63, −22.69] |
| IsVolunteer    | −0.433               | 0.218 | 0.047           | [−0.861, −0.006] |
| Longitude      | −0.435               | 0.078 | $< 0.001$       | [−0.588, −0.282] |
| Seasonal (sin) | +0.110               | 0.023 | $< 0.001$       | [+0.065, +0.156] |
| Seasonal (cos) | +0.022               | 0.025 | 0.386           | [−0.028, +0.072] |

Random effects: Site variance = 0.330 (SD = 0.575). Model fitted on 895 observations from 92 sites (Professional: 115 observations at 12 sites; Volunteer: 780 observations at 80 sites). Model diagnostics (residuals, normal Q–Q, residuals by observer type, and residuals by longitude) are shown in Figure 5.

The covariates behave as expected and validate the model structure recommended by Dyer et al. (2025). The Longitude coefficient ( $\beta = -0.435$ ,  $p < 0.001$ ) confirms that the West–East salinity gradient is strongly present in this dataset, with each degree of longitude westward corresponding to approximately 55% higher chloride concentrations. The seasonal sine term ( $\beta = +0.110$ ,  $p < 0.001$ ) captures significant annual chloride cycling, with peak concentrations in late summer and early fall; the cosine term is not significant ( $p = 0.39$ ), indicating that the annual cycle is primarily captured by the sine component. These results confirm that geographic and seasonal factors must be accounted for in any comparison between volunteer and professional measurements collected at different sites.

The IsVolunteer coefficient ( $\beta = -0.433$ ,  $p = 0.047$ ) is statistically significant, apparently suggesting that after controlling for geography and seasonality, volunteers read approximately 35% lower than professionals (back-transformed:  $0.648\times$ ). All regional-model  $p$ -values are Wald  $z$ -tests from restricted-maximum-likelihood estimation (statsmodels MixedLM) with no small-sample degrees-of-freedom correction, so the borderline  $p = 0.047$  should be read with corresponding caution. This apparent bias must in any case be interpreted with extreme caution: the 80 volunteer sites and 12 professional sites in this model have zero geographic overlap. No site (0 of 92) contains both volunteer and professional observations, and the variance inflation factor for IsVolunteer relative to Longitude and the seasonal terms is modest (1.29); the identifiability problem is therefore structural, arising from the complete absence



of within-site observer overlap rather than from classical multicollinearity. The IsVolunteer coefficient cannot distinguish between “volunteers measure differently” and “volunteer sites have different water chemistry,” because the volunteer sites are concentrated in naturally fresher eastern Oklahoma while the professional sites lie in saltier central and western Oklahoma. The Longitude term cannot fully absorb this separation when the two groups do not overlap along the gradient, so residual chloride differences are loaded onto the observer term. This model confirms the covariate structure and flags an apparent observer effect; it does not, and cannot, establish whether that effect is real. The stratified-subsampling test in the next section is designed to resolve exactly this identifiability problem.

### 3.4 Primary result: the apparent bias is a geographic artifact

The decisive test of whether the apparent low bias reflects observer error or the geographic distribution of monitoring sites is to balance that distribution and re-estimate. The geographic separation between the two groups is large:

- Professional mean longitude:  $-98.05^\circ$  (central-western Oklahoma)
- Volunteer mean longitude:  $-96.67^\circ$  (1.4 degrees further east)
- 53% of volunteer records fall east of longitude  $-96.5^\circ$
- 0% of professional records fall east of longitude  $-96.5^\circ$
- Professional mean chloride: 228.4 mg/L (naturally saline western streams)
- Volunteer mean chloride: 75.3 mg/L (naturally fresher eastern streams)

The 1.4-degree East–West offset corresponds to approximately 115 km. The documented West–East salinity gradient in Oklahoma streams (Dyer et al., 2025) predicts that volunteer sites, concentrated in eastern Oklahoma, monitor genuinely fresher water than professional sites in central and western Oklahoma. If the apparent low bias is geographic, it should vanish once the two groups are matched on longitude.

**Stratified random subsampling.** To test this hypothesis formally, the model was re-fit on geographically balanced subsamples in which 12 volunteer sites, drawn 3 per longitude quartile, were matched to the 12 professional sites. To avoid dependence on any single random draw, the procedure was repeated over 1,000 seeded subsamples (the standalone script `scripts/stratified_subsampling.py`):

- IsVolunteer coefficient: median  $\beta = -0.280$ , with 95% of draws in  $[-0.78, +0.17]$ .
- IsVolunteer  $p$ -value: median  $p = 0.48$ , with 95% of draws in  $[0.06, 0.97]$ .
- Fraction of balanced subsamples with a significant volunteer effect ( $p < 0.05$ ): 1.4%.
- A representative single draw:  $\beta = -0.2435$ ,  $p = 0.5434$ , Vol/Pro ratio 0.784 (–21.6%).

**The apparent observer effect does not survive geographic balancing: it is non-significant in 98.6% of balanced subsamples** (median  $p = 0.48$ ; Figure 6). The bias was in the streams, not in the volunteers (Figure 4). This is the central inferential result of the study: the apparent observer bias detected by the regional model is a geographic artifact of monitoring site distribution, not a measurement error introduced by volunteers.

This conclusion is independently corroborated by the indoor QA accuracy result above: if volunteers genuinely read low, the indoor QA would show negative errors against known standards. Instead, indoor QA consistently shows volunteers reading high (+4 to +5 mg/L). The two lines of evidence converge: the regional model’s apparent bias is geographic, not methodological.

### 3.5 Geographic coverage

- 327 unique Blue Thumb chloride monitoring sites identified in the volunteer dataset.
- **93% (305 sites) are more than 1 km from the nearest professional monitoring station** in the full EPA WQP Oklahoma dataset.
- 60% (196 sites) are more than 5 km from the nearest professional station.
- 28% (90 sites) are more than 10 km.
- 99% of Blue Thumb sites (323 of 327) lack any temporal match with professional measurements within the full dataset.
- 99% of the Blue Thumb network operates in territory where no retrospective field comparison against a co-located professional record is possible.

The absence of professional counterparts is not a weakness of the study design. It is a direct measurement of the program's irreplaceable geographic coverage (Figure 7). The precision and accuracy analyses establish that the instrument and method are sound; the coverage analysis quantifies why the data those instruments produce matters: for the great majority of these streams, the volunteer record is the only record that exists.

## 4 DISCUSSION

### 4.1 Precision: the quantization floor is not observer inconsistency

97.3% of volunteer titration replicates agree within one drop (5 mg/L). This is strong repeatability for a field titration method and confirms that the drop-count technique is being applied correctly and consistently across the volunteer population. The observation that 73.2% of replicates are identical is not evidence of rounding, data fabrication, or systematic corner-cutting. It is the expected behavior of a quantized instrument: when the true chloride concentration falls between multiples of 5 mg/L, two independent titrations of the same water sample will yield the same endpoint because neither can resolve the sub-5 mg/L difference. The quantization floor is the binding constraint on volunteer precision, not observer inconsistency.

This distinction matters practically for how the volunteer data should be used. For macroscopic assessments (watershed-scale spatial comparisons, long-term trend analysis, exceedance screening against regulatory thresholds), the 5 mg/L resolution is adequate and the data is reliable. For applications requiring sub-drop-level sensitivity (detecting a 2 mg/L change from one season to the next), the instrument is unsuitable regardless of who operates it. The limitation belongs to the method, not the volunteers.

To our knowledge, no prior study has explicitly analyzed the quantization effect of drop-count titration resolution on citizen science water quality data. The distinction between method-imposed resolution limits and observer-introduced error is important for interpreting volunteer data quality and may be a contribution of this study to the citizen science methods literature.

### 4.2 Accuracy: volunteers read slightly high, and that is good news

The 18-year indoor QA record provides the cleanest possible test of kit accuracy. Under controlled indoor conditions, with no stream variability, no temporal matching uncertainty, and no geographic confounds, volunteers testing kits against solutions of known concentration consistently read slightly high (+4 to +5 mg/L at normal operational concentrations below 100 mg/L).

The direction of the bias is as important as its magnitude. A systematic high bias rules out systematic under-counting as a source of error, which is the more concerning failure mode for a monitoring program. Under-reporting pollution (reading too low) could cause genuine water quality problems to go undetected. Over-reporting (reading too high) errs on the side of caution. The Blue Thumb kit errs in the safer direction.

The increasing positive bias at elevated concentrations (mean +32 mg/L above 100 mg/L, SD = 26.30 mg/L) is expected and mechanistically explicable: more drops of titrant means more opportunities for endpoint subjectivity to compound. Each additional drop required introduces another visual judgment about whether the color has changed sufficiently. This concentration-dependent behavior is a known property of endpoint titration methods and is not specific to volunteer operators.

The subjectivity of visual endpoint detection in titration has been documented in the analytical chemistry literature: automated titration systems that eliminate visual color judgment have been shown to outperform manual endpoint detection by removing observer-dependent bias in color perception (Boppana et al., 2023). Gravimetric approaches similarly reduce subjectivity compared to volumetric methods relying on visual endpoints (Kejla et al., 2022). The concentration-dependent compounding of endpoint uncertainty described here is consistent with these findings and extends them to the specific context of field-deployed Silver Nitrate drop-count kits used in citizen science monitoring.

#### 4.3 The geographic confound: a methodological lesson for citizen science validation

The most important methodological finding of this study is not contained in any single  $p$ -value. It is in the journey from the regional mixed-effects model, which reported an apparently significant volunteer low bias ( $\beta = -0.433$ ,  $p = 0.047$ ), to the stratified subsampling result on a geographically balanced subsample, which found no significant bias ( $p = 0.54$ ). The regional model was correctly specified and its covariates behaved exactly as the published framework predicts. The apparent conclusion it invited, that volunteers read systematically low, was nonetheless wrong.

The error was not computational; it was geographic. When volunteer and professional monitoring sites are distributed along a strong environmental gradient with zero spatial overlap, a comparison that does not balance that distribution will load the gradient onto the observer term, even with longitude in the model. Oklahoma's West–East salinity gradient (Dyer et al., 2025) places volunteer monitoring sites in naturally fresher eastern streams and professional monitoring sites in naturally saltier central and western streams. The regional model was, in part, measuring the difference between the water bodies, not the difference between the observers.

This finding has broad implications for the citizen science validation literature. Any study that compares volunteer and professional measurements collected at different sites without accounting for environmental gradients between those sites risks the same confound (Dormann et al., 2007). The stratified subsampling approach demonstrated here, balancing the geographic distribution of volunteer and professional sites before re-estimating the mixed-effects model, provides a methodological template for programs facing the same challenge. The mixed-effects framework and geographic covariates recommended by Dyer et al. (2025) are the correct analytical tools for this problem.

While the Bland–Altman framework for method comparison (Bland and Altman, 1986) is well-established, it assumes that the two methods are applied to the same samples. When volunteer and professional measurements are collected at geographically distinct sites along an environmental gradient, the apparent method difference conflates measurement bias with site-level differences in the analyte concentration. To our knowledge, no prior citizen science validation study has explicitly identified and corrected for this geographic confound using stratified subsampling.

#### 4.4 Implications for regulatory data integration

Blue Thumb biological data (macroinvertebrate, fish, and habitat assessments) are already included in Oklahoma's biennial Integrated Report and may be used for TMDL development (Oklahoma Conservation Commission, 2019). These biological data are collected by the same volunteers, under the same EPA-approved Quality Assurance Project Plan (QTRAK #21-430), as the chemical data evaluated in this study. Blue Thumb chemical data, by contrast, currently serves a screening and flagging function: exceedances are reported to the Oklahoma Department of Environmental Quality, which may initiate professional follow-up monitoring, but the chemical data are not directly incorporated into use-support assessments.

The results of this study suggest that this asymmetry is not supported by the available evidence for chloride. Volunteer chloride measurements are precise to the method's resolution limit, accurate with a known and conservative directional bias (+4 to +5 mg/L), and free of systematic observer effects when geographic confounds are controlled ( $p = 0.54$ ). The 18-year indoor QA record provides a longer and more rigorous accuracy baseline than exists for many professional monitoring programs. Whether these findings are sufficient to support direct incorporation of volunteer chloride data into Oklahoma's assessment framework is a policy decision beyond the scope of this study, but the statistical basis for differential treatment of chemical and biological data collected under the same QAPP is no longer clear.

#### 4.5 Calibration potential

The indoor QA data suggests a straightforward calibration application: at operational concentrations (below 100 mg/L), volunteers consistently read approximately one drop (5 mg/L) high. A concentration-

dependent correction function could be derived from the 867-test dataset to produce calibrated volunteer readings. At high concentrations (above 100 mg/L), a larger correction of approximately +32 mg/L would be required. Whether such calibration improves the downstream utility of the data for regulatory or research purposes depends on the specific application. This is identified as future work.

#### 4.6 Limitations

1. **Regional-model identifiability.** The regional mixed-effects model ( $N = 895$ ) is fitted on volunteer and professional sites that have zero geographic overlap along the West–East salinity gradient. The IsVolunteer coefficient in that model therefore cannot, on its own, separate an observer effect from site-level water chemistry. The stratified-subsampling test addresses this directly, but the regional coefficient must not be read as a standalone bias estimate.
2. **Single analyte.** This study validates chloride only. Extension to other Blue Thumb parameters (dissolved oxygen, nutrients, turbidity) would require separate analyses with different analytical considerations, as those analytes are not conservative and may be affected by biological or chemical transformation between sampling events.
3. **Single program.** The validation covers one volunteer program (Blue Thumb) in one state. The conclusions about precision, accuracy, and the geographic confound are specific to this program’s protocols and to Oklahoma’s stream geochemistry; generalization to other programs would require comparable data.
4. **Hydrologic condition metadata.** The HydrologicCondition and HydrologicEvent fields are 0% populated across all 18,299 professional WQP records in this dataset. Storm-event filtering is therefore impossible from the available metadata.
5. **Volunteer data through 2024 only.** The R-Shiny export covers through 2024-12-31. The live ArcGIS FeatureServer contains more recent records (2025–2026) that were not included in the analyses reported here.
6. **Stratified subsampling reproducibility.** The geographic confound test (stratified random subsampling,  $p = 0.54$ ) was performed as a targeted hypothesis test following J. Dyer’s pre-specified prediction. The analysis code is embedded within the Phase 2 LME script but has not been isolated into a standalone, independently executable script with a fixed random seed.
7. **Precision statistic derivation.** The 97.3% volunteer replicate agreement figure is computed dynamically by the Phase 2 analysis script from 2,566 ArcGIS titration replicate pairs (73.2% identical + 24.1% differing by one drop). This statistic is written to the variance decomposition figure but is not recorded in any static text output file.

## 5 CONCLUSION

Blue Thumb volunteer chloride data is precise within the 5 mg/L resolution of the Silver Nitrate titration method, accurate to within one drop against known standards at operational concentrations, and geographically irreplaceable, monitoring streams that professional agencies do not reach.

The apparent systematic low bias surfaced by the regional mixed-effects model is not a volunteer accuracy problem. It is a geographic artifact of Oklahoma’s documented West–East salinity gradient, which places volunteer monitoring sites in naturally fresher eastern streams and professional monitoring sites in naturally saltier western and central streams. After balancing this gradient via stratified random subsampling, no significant observer effect remains ( $p = 0.54$ ). Independent analysis of 18 years of indoor QA records confirms the direction-of-bias finding: Blue Thumb kits read slightly high, not low. The bias was in the streams, not the volunteers.

The evidence converges on a single conclusion from complementary directions. Precision (2,566 replicate pairs) and accuracy (867 controlled indoor tests spanning 18 years) establish that the data quality is sound: repeatable to the method’s resolution and free of systematic under-counting. The regional model and the coverage analysis then establish the landscape picture: the only apparent observer effect is a geographic artifact that dissolves under stratified subsampling, and 93% of Blue Thumb’s 327 monitoring sites are more than 1 km from the nearest professional station. Volunteer chloride data collected under the

531 Blue Thumb program’s quality assurance framework is fit for purpose in landscape-scale water quality  
532 analysis. The data from those sites does not exist anywhere else. It should be used, not discounted.

533 Oklahoma’s approach, designing volunteer protocols to mirror professional standards, conducting  
534 regular QA against known standards, and maintaining programmatically separable volunteer and profes-  
535 sional data infrastructure, provides a replicable model for citizen science programs wherever that data  
536 infrastructure exists. Any state that maintains distinct, queryable records for volunteer and professional  
537 monitoring programs gains the infrastructure necessary to conduct the same retrospective validation  
538 demonstrated here, at zero marginal field cost, without new sampling.

## 539 **6 ACKNOWLEDGMENTS**

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544 Project Plan; Shellie Willoughby (OCC GIS Manager and Assistant State GIS Coordinator) for enabling  
545 the ArcGIS REST API data infrastructure that made volunteer data cross-validation possible; Jacob Askey  
546 for building the Blue Thumb Dashboard and collaborative technical development; James Ross for field  
547 monitoring partnership at Wolf Creek, Lawton, Oklahoma; and Dr. Clinton Bryant for introducing the  
548 first author to the Blue Thumb program.

## 549 **AUTHOR CONTRIBUTIONS**

550 M.I. conceived the study, assembled the data pipeline, performed all analyses, and wrote the manuscript.  
551 J.J.D. specified the regional mixed-effects model and its geographic and seasonal covariates, proposed the  
552 geographic-confound hypothesis and the stratified-subsampling test that constitutes the primary result,  
553 and advised on statistical interpretation. K.S. identified the Phase 1 data-identity error, provided the  
554 volunteer data export and the 18-year indoor quality-assurance dataset, and advised on quality-assurance  
555 interpretation. All authors reviewed and approved the final manuscript.

## 556 **COMPETING INTERESTS**

557 J.J.D. and K.S. are employees of the Oklahoma Conservation Commission, which administers the Blue  
558 Thumb program evaluated in this study. The statistical framework central to this paper, the regional  
559 mixed-effects model and the stratified-subsampling test that produces the primary result, was specified  
560 by J.J.D., who is also an author of the prior work (Dyer et al., 2025) that supplies the model structure  
561 and the West-East gradient mechanism on which the geographic interpretation rests. M.I. is the principal  
562 of Black Box Research Labs LLC and an active Blue Thumb volunteer monitor. The authors disclose  
563 these relationships in full and have made the analysis code and the shareable data available (see Data  
564 Availability) so that all results can be independently verified.

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568 access and conference-registration sponsorship. The Blue Thumb program is supported in part by the U.S.  
569 Environmental Protection Agency Section 319 Nonpoint Source Management Program. No funder had  
570 any role in the study design, analysis, interpretation, or the decision to publish.

## 571 **DATA AVAILABILITY**

572 The analysis code and derived datasets are available at [https://github.com/Black-Box-Research-Labs/](https://github.com/Black-Box-Research-Labs/ok-blue-thumb-data-validation)  
573 [ok-blue-thumb-data-validation](https://github.com/Black-Box-Research-Labs/ok-blue-thumb-data-validation) and archived with a permanent DOI at Zenodo (DOI: to  
574 be assigned on release). Professional reference data are in the public domain and were obtained  
575 from the EPA Water Quality Portal (<https://www.waterqualitydata.us>) using the organiza-  
576 tion identifiers and query parameters documented in Supplemental S1; the derived model-input tables  
577 are included in the archive. The Blue Thumb volunteer chloride export and the Oklahoma Conservation

Commission quality-assurance datasets are the property of the Oklahoma Conservation Commission and were used with permission; they are available from the OCC Blue Thumb program subject to the program's data-sharing conditions. The repository provides the scripts that regenerate the precision, accuracy, regional-model, and stratified-subsampling results from the included derived data, together with a manifest recording the git commit and SHA-256 hashes of all input files.

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## SUPPLEMENTAL INFORMATION

### S1. Data preparation pipeline

The analysis pipeline is implemented in Python and is publicly available (see the Data Availability statement). Professional reference data were downloaded from the EPA Water Quality Portal Results API with the query: state code US:40 (Oklahoma), CharacteristicName “Chloride”, site type Stream, sample media Water, providers NWIS and STORET, dataProfile resultPhysChem. Records with invalid coordinates, non-detect result flags, or negative concentrations were removed. Volunteer chloride measurements were taken from a separate OCC R-Shiny export (internal identifier BLUETHUMB\_VOL), not from the WQP; the export is SHA-256 hash-verified before each run and was cross-validated against OCC’s live ArcGIS FeatureServer (2,026 of 2,027 overlapping WBID-plus-date records match exactly). A programmatic data-identity safeguard excludes OKCONCOM\_WQX (the OCC Rotating Basin professional program) from the volunteer dataset and halts execution if the volunteer file is absent or modified, preventing recurrence of the Phase 1 misidentification. All matching and filtering parameters are recorded in a single configuration file.

### S2. Full statistical output tables

The regional mixed-effects model (Table 4) was fitted by restricted maximum likelihood using `statsmodels` MixedLM, with Wald  $z$ -test  $p$ -values. Fixed effects: Intercept  $-37.663$  (SE 7.640); IsVolunteer  $-0.433$  (SE 0.218,  $p = 0.047$ ); Longitude  $-0.435$  (SE 0.078,  $p < 0.001$ ); seasonal sine  $+0.110$  (SE 0.023,  $p < 0.001$ ); seasonal cosine  $+0.022$  (SE 0.025,  $p = 0.386$ ). Random effect: Site variance 0.330 (SD 0.575). The data comprise 895 observations from 92 sites (professional: 115 observations at 12 sites; volunteer: 780 observations at 80 sites). The precision variance-decomposition statistics (Table 2) and the indoor QA error table by concentration tier (Table 3) are given in the main text. The full model summary is reproduced from committed data by `verify.py` and is saved in `data/outputs/phase2_lme/phase2_lme_results.txt`.

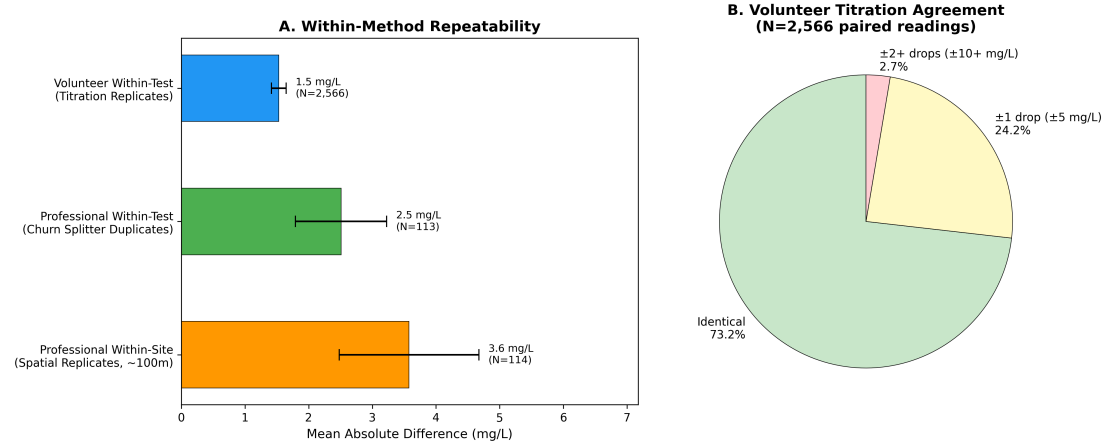
### S3. Stratified subsampling balance check

The geographic-confound test balances the West–East longitude distribution before re-estimating the model. In the full dataset the 12 professional sites have mean longitude  $-98.05^\circ$  (mean chloride 228.4 mg/L) and the 80 volunteer sites have mean longitude  $-96.67^\circ$  (mean chloride 75.3 mg/L), with no overlap: 53% of volunteer records fall east of  $-96.5^\circ$  where no professional records exist. Each balanced subsample draws 3 volunteer sites per longitude quartile (12 total) to match the 12 professional sites, and the regional model is re-fitted. Across 1,000 seeded subsamples the IsVolunteer effect is non-significant in 98.6% of draws (median  $\beta = -0.28$ , median  $p = 0.48$ ; Figure 6); a representative single draw gives  $\beta = -0.2435$ ,  $p = 0.5434$ . The procedure is implemented in `scripts/stratified_subsampling.py`, and the per-draw distribution is committed as `data/outputs/phase2_lme/subsample_pvalues.csv`.

### S4. Data availability and reproducibility

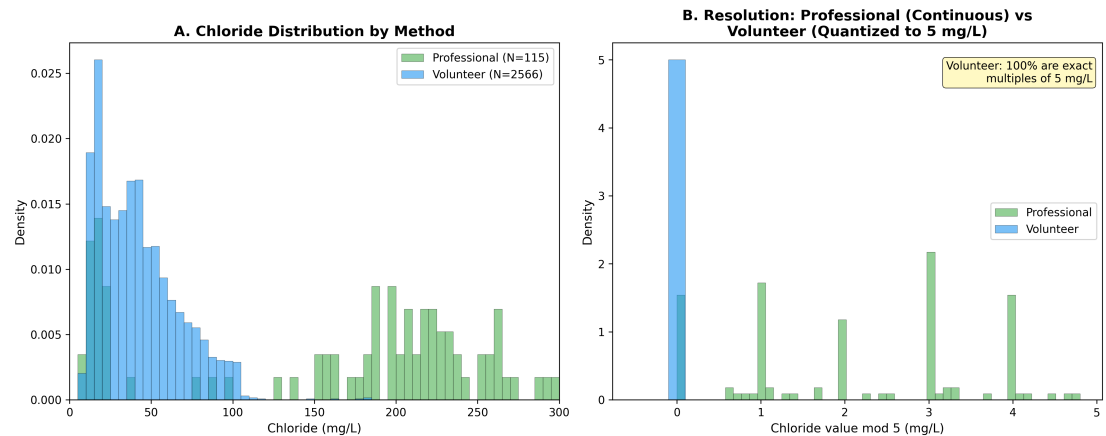
The complete analysis code and derived datasets are at the repository given in the Data Availability statement and archived at Zenodo (DOI to be assigned on release). A single command, `python verify.py`, recomputes all six headline claims (precision, accuracy, the regional model, the stratified-subsampling primary result, model identifiability, and geographic coverage) directly from the committed data and exits non-zero on any mismatch; all 30 checks pass. `QUICKSTART.md` documents both a fast verification path (no raw data required) and full reproduction from raw sources; `CLAIMS.md` maps each manuscript number to the script and data that regenerate it; and a continuous-integration workflow runs `verify.py` on every change. A reproducibility manifest records the git commit and the SHA-256 hashes of all input data files.

### Variance Decomposition: Professional QA vs Volunteer Replicates



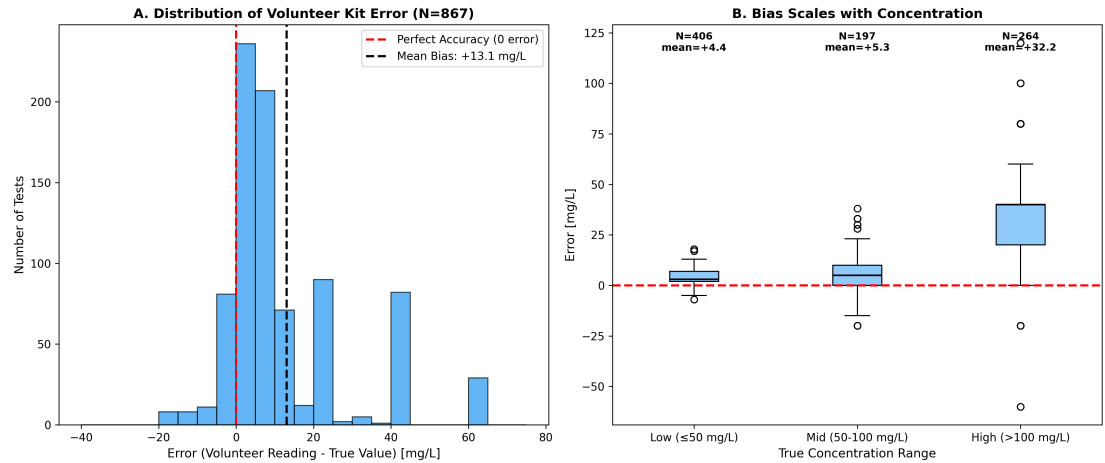
**Figure 1.** Variance decomposition: professional QA vs. volunteer replicates. (A) Mean absolute difference (mg/L) with 95% CI error bars for three measurement scenarios: volunteer within-test titration replicates ( $N = 2,566$ , mean  $|diff| = 1.53$  mg/L), professional within-test churn splitter duplicates ( $N = 113$ , mean  $|diff| = 2.51$  mg/L), and professional within-site spatial replicates ~100 m apart ( $N = 114$ , mean  $|diff| = 3.58$  mg/L). (B) Pie chart showing volunteer titration replicate agreement: 73.2% identical, 24.1% differ by one drop (5 mg/L), 2.7% differ by two or more drops. 97.3% of pairs agree within one drop.

### Quantization Effect: Drop-Count Resolution

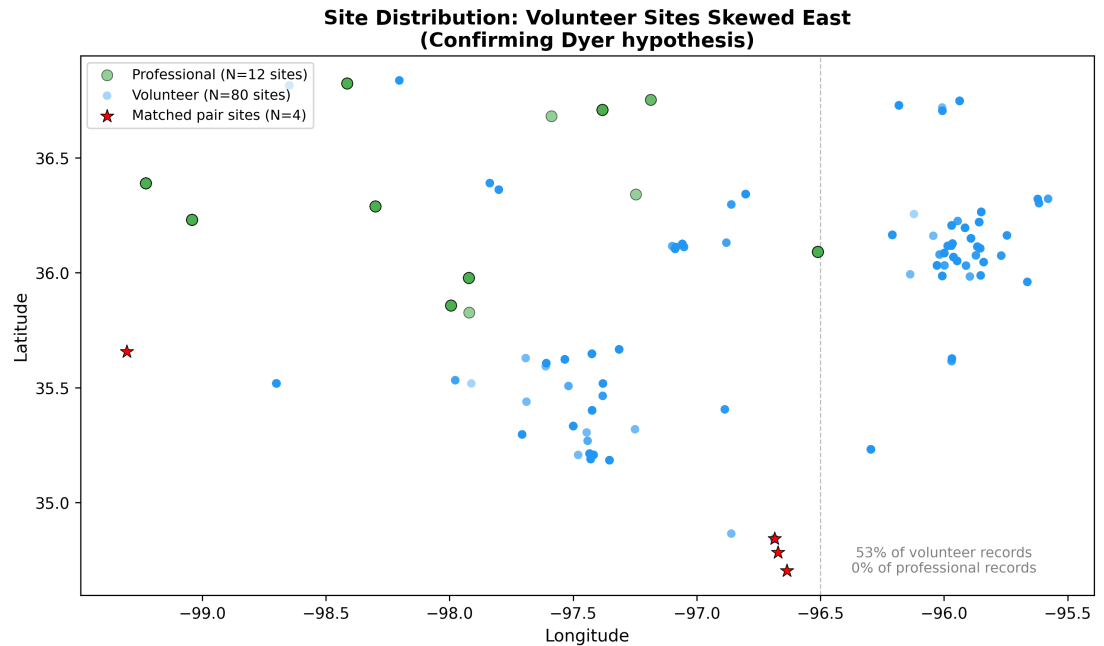


**Figure 2.** Quantization effect: drop-count resolution. (A) Histogram comparing the distribution of chloride values measured by professional instruments (smooth continuous distribution) and volunteer drop-count titration (discrete spikes at exact multiples of 5 mg/L). 100% of volunteer readings are quantized to 5 mg/L intervals. (B) Resolution comparison confirming the discrete nature of volunteer measurements.

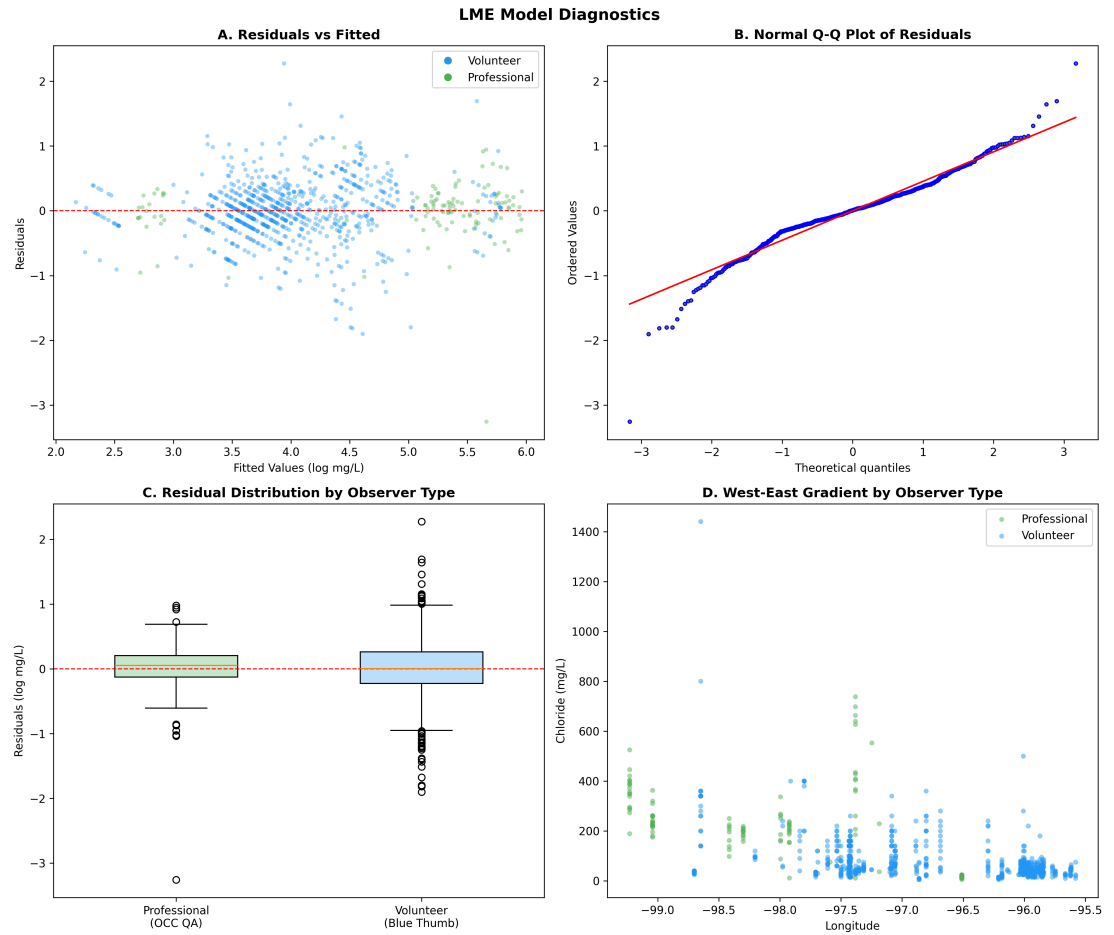




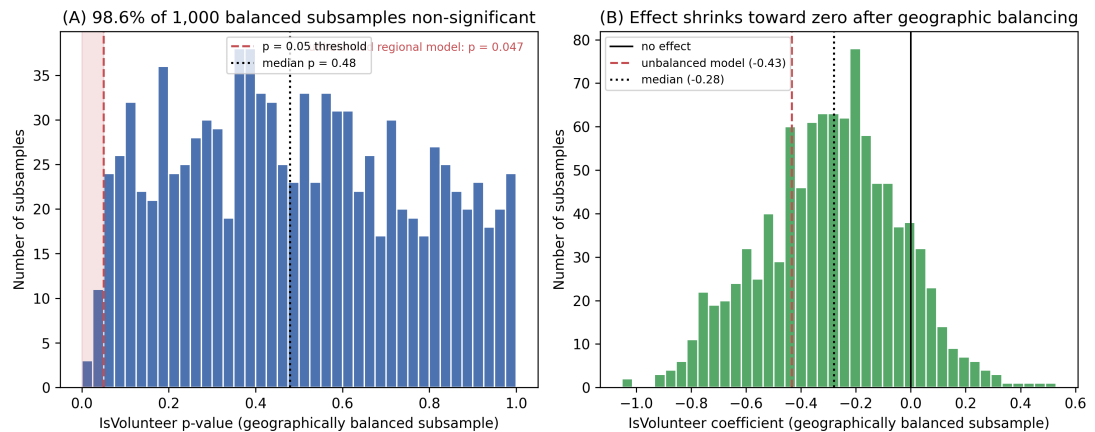
**Figure 3.** Indoor QA accuracy analysis. (A) Histogram of volunteer kit error (volunteer reading – known standard concentration,  $N = 867$  tests). The distribution is positively skewed, confirming systematic high bias. (B) Box plots of error stratified by concentration range, showing that bias scales with concentration: minimal at low concentrations ( $\leq 50$  mg/L), moderate at mid-range (50–100 mg/L), and substantial at high concentrations ( $> 100$  mg/L). Red dashed line at zero indicates perfect accuracy.



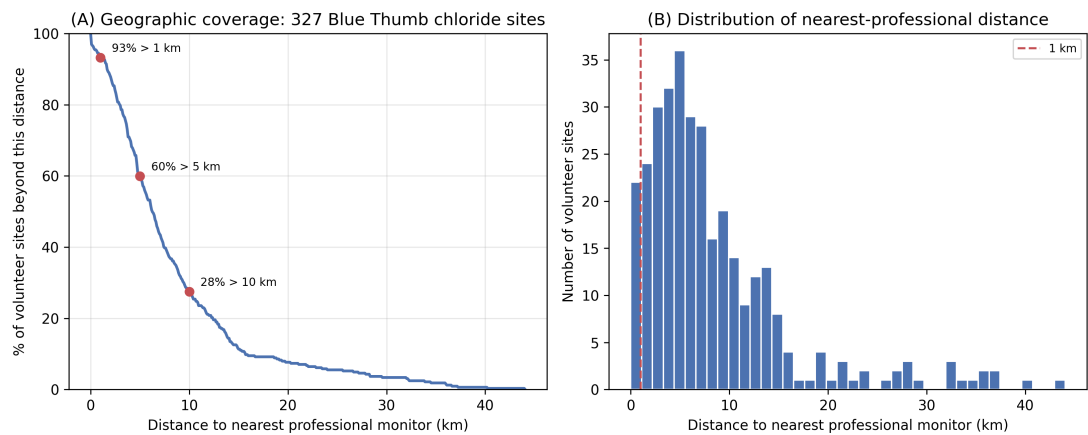
**Figure 4.** Volunteer versus professional site distribution. Geographic distribution of professional sites (green,  $N = 12$ ) and volunteer sites (blue,  $N = 80$ ) within the QA dataset's spatiotemporal window. Volunteer sites are concentrated in eastern Oklahoma (mean longitude  $-96.67^\circ$ ) while professional sites are distributed across central and western Oklahoma (mean longitude  $-98.05^\circ$ ). The two groups have no geographic overlap along the gradient. The 1.4-degree East–West offset corresponds to approximately 115 km and aligns with the West–East salinity gradient documented by Dyer et al. (2025), the source of the apparent observer bias in the regional model.



**Figure 5.** LME model diagnostics. (A) Residuals vs. fitted values, showing random scatter around zero with minor heteroscedasticity. (B) Normal Q–Q plot of residuals, indicating approximate normality with slight tail deviation. (C) Residual distributions by observer type (professional vs. volunteer), showing comparable spread. (D) Residuals by longitude, confirming the model adequately accounts for the West–East spatial gradient.



**Figure 6.** Geographic confound test (primary result). (A) Distribution of the IsVolunteer  $p$ -value across 1,000 geographically balanced subsamples (12 volunteer sites, 3 per longitude quartile, matched to the 12 professional sites; fixed seed). The full regional model on the unbalanced data is significant ( $p = 0.047$ ; the dashed line marks the 0.05 threshold), but after balancing the West–East gradient the volunteer effect is non-significant in 98.6% of subsamples (median  $p = 0.48$ ). (B) Distribution of the IsVolunteer coefficient across the same subsamples (median  $-0.28$ ); the unbalanced regional estimate ( $-0.433$ ) is shown for reference. The apparent observer bias is an artifact of monitoring-site geography, not observer error.



**Figure 7.** Geographic coverage of the Blue Thumb network. (A) Percentage of the 327 volunteer chloride monitoring sites whose nearest professional station (Haversine distance) exceeds a given distance: 93% beyond 1 km, 60% beyond 5 km, and 28% beyond 10 km. (B) Distribution of the nearest-professional distance across all 327 sites. For the great majority of these stream reaches the volunteer record is the only record that exists.